Accurate inference of genetic ancestry from cancer-derived data

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ABSTRACT

INTRODUCTION: Genetic ancestry-oriented cancer research requires the ability to **perform accurate and robust genetic ancestry inference from existing cancer-derived data**, including whole exomes, transcriptomes and targeted gene panels, very often **in the absence of matching cancer-free genomic data**. Here we examine the feasibility and accuracy of such computation. In order to optimize and assess the performance of the ancestry inference for any given input cancer-derived molecular profile, we have developed a data synthesis framework.

METHODS: In its core procedure, the ancestral background of the profiled patient is replaced with one of any number of individuals with known ancestry. Data synthesis is applicable to multiple profiling platforms and makes it possible to assess the performance of inference specifically for a given molecular profile, and separately for each continental-level ancestry. This ability extends to all ancestries, including those without statistically sufficient representation in the existing cancer data.

IMPLEMENTATION: The inference procedure is implemented as an R package called Robust Ancestry Inference using Data Synthesis (RAIDS). The package uses the genomic data structures and methods provided by the Bioconductor GenomicRanges, SNPRelate, and gdsfmt packages. The data synthesis is part of the RAIDS package.

RESULTS: We show that our inference procedure is accurate and robust in a wide range of sequencing depths. Testing our approach for three representative cancer types, and across three molecular profiling modalities, we demonstrate that global, continental-level ancestry of the patient can be inferred with high accuracy, as quantified by its agreement with the golden standard of the ancestry derived from matching cancer-free molecular data.

CONCLUSION: Our study demonstrates that vast amounts of existing cancer-derived molecular data potentially are amenable to ancestry-oriented studies of the disease, without recourse to matching cancer-free genomes or patients' self-identification by ancestry. Computational tools for this purpose are available as open-source software.

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Ancestry inference from cancer-derived data

Aim: Reliably infer genetic ancestry from cancer-derived genomic data

- We have developed a pipeline to infer the ancestry of cancer samples [1] (**Figure 1**)
- Pipeline to generate synthetic data from cancer-derived data for parameter optimization and performance assessment (**Figure 2**)
- Pipeline to infer ancestry from cancer-derived genomic data (**Figure 3**)
- Description of the 3 datasets used to illustrate the method (**Figure 4**)
- Results from the synthetic data, illustration based on pancreatic cancer dataset to generate American (AMR) synthetic data (**Figure 5**)
- Performance of the ancestry inference by comparison to the ancestry inferred from the matching cancer-free data (**Table 1**)
- Confusion matrix comparing the cancer-free ancestry inference to the ancestry inference from cancer-derived data (**Table 2**)

Methods

Overview

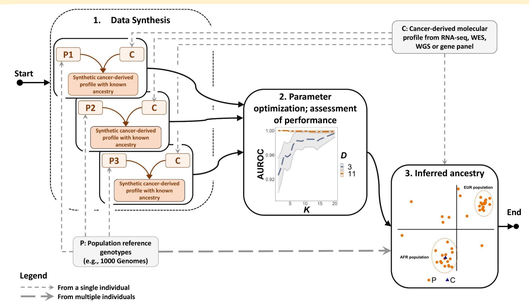

Figure 1. An overview of genetic ancestry inference from cancer-derived molecular data using data synthesis.

Steps:

- We start with a cancer dataset (C).
- We used the cancer profile information such as sequencing coverage and allelic fraction to generate a synthetic dataset with the same properties and the genotype from different reference samples (P1, P2, …) and different populations.
- Based on the synthetic data, we estimate the best parameters to analyze this profile.
- We infer the ancestry based on those parameters.

Abbreviations: Whole exome sequencing (WES), Whole genome sequencing (WGS)

Synthetic data

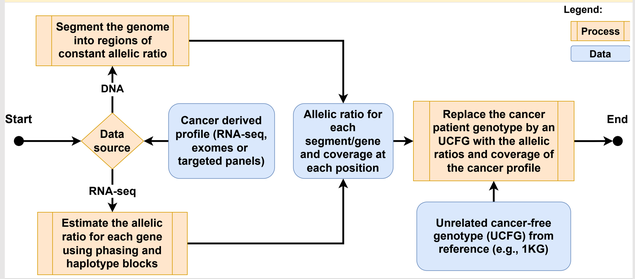

Figure 2. A schematic overview of the data synthesis process. Data synthesis is defined here as replacement of the sequence variants detected in a cancer-derived profile P by those found in the genome of an unrelated individual U. Ingredients required for this procedure are: (a) allele fraction (AF) estimates in P and (b) the haplotype of U in the portion of the genome covered by P. With this knowledge, the procedureconsists of the following steps. First, sequence reads comprising P are distributed at random among the alleles with probabilities equal to the observed allele fractions. Second, in each haplotype block in the genome of U that is covered by P, allele assignment is made at random, yielding variant and reference read counts for each substitution in the genome of U within the scope of P.

Inference

- 1000 Genomes (1KG) as the reference data set
- 5 ancestral super-populations: East Asian (EAS), European (EUR), African (AFR), American [Hispanic] (AMR), South Asian (SAS)
- Retain single-nucleotide variants (SNV) frequent in at least one super-population

For each sample:

1. Intersect with SNV positions covered above minimal depth in the WES from the sample
2. Use these as a basis for principal component analysis [2]
3. K-nearest neighbors classification in the subspace of 1st D principal components

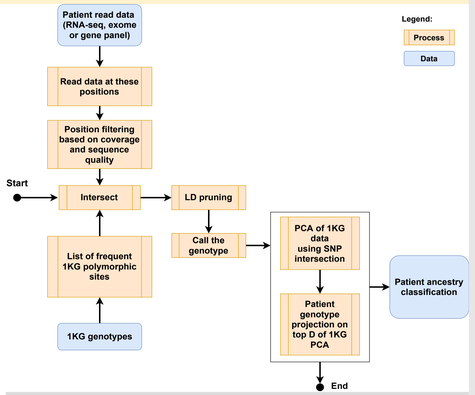

Figure 3. A flowchart of the inference of genetic ancestry.

Dataset

Summary of the molecular data used in this study (**Figure 4**). These originate from three patient cohorts:
A) TCGA ovarian cancer [3]
B) acute myeloid leukemia [4]
C) pancreatic ductal adenocarcinoma (PDAC) library of patient-derived organoids [5]

D) The distribution of the patients, as self-declared, for Beat AML, PDAC and TCGA-OV cohorts. UNK means not reported or unknown.

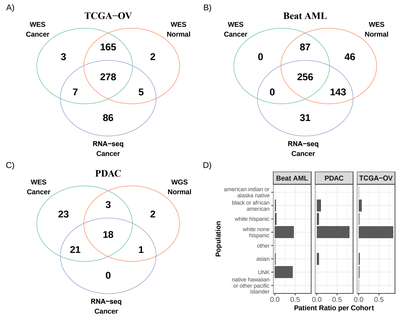

Figure 4. Dataset used in the study.

Results – Synthetic data

Dependence of AMR-specific AUROC* on the inference parameters D and K, computed using data synthesis for 10 PDAC patients and the three profiling modalities: WES, RNA-seq and FoundationOne® CDx Panels**(**Figure 5**). The central AUROC values are shown in solid lines and the 95% CI in dashed lines.

* Area under the receiver operating characteristic curve (AUROC) [6]. AUROC is a measure of the call.

** The real data are not available for FoundationOne® CDx Panels. We used exome data restricted to the genes panel.

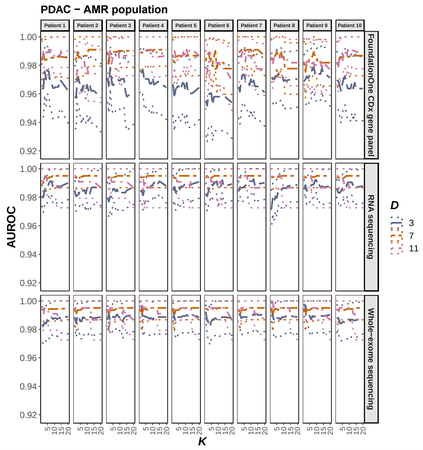

Figure 5. Synthetic data based on PDAC profiles to generate American (AMR) data.

Results – Real data

Table 1. Cohort-wide performance measures for super-population calls from cancer-derived molecular data. A reliable estimate of the confidence intervals (CI) was not possible in the case of PDAC, due to the small number of cases with matching cancer-free genotypes.

Study	D	K	Accuracy	95% CI	AUROC	95% CI
TCGA-OV WES	5	13	0.998	0.994-1	0.993	0.992-0.994
TCGA-OV Panel	4	12	0.984	0.972-0.996	0.966	0.965-0.967
TCGA-OV RNA-seq	7	12	0.993	0.983-1	0.977	0.975-0.979
BeatAML WES	5	13	0.989	0.978-1	0.978	0.976-0.980
BeatAML Panel	4	13	0.991	0.981-1	0.999	0.999-0.999
BeatAML RNA-seq	4	13	0.992	0.981-1	0.999	0.999-0.999
PDAC WES	8	13	1	NA	NA	NA
PDAC Panel	6	5	0.952	0.861-1	0.958	NA
PDAC RNA-seq	4	13	1	NA	NA	NA

Table 2. Confusion matrices comparing TCGA-OV or Beat AML patients' super-population calls from the three different cancer-derived molecular sources (WES, FoundationOne® CDx Panel and RNA-seq) to the calls from the matching cancer-free WES.

(a) TCGA-OV WES

		Inferred				
	pop	EAS	EUR	AFR	AMR	SAS
Cancer-free WES	EAS	10	0	0	0	0
	EUR	0	378	0	0	0
	AFR	0	0	29	0	0
	AMR	0	1	0	16	0
	SAS	0	0	0	0	7
	UNK	0	2	0	0	0

(b) BeatAML WES

	Inferred				
pop	EAS	EUR	AFR	AMR	SAS
EAS	11	0	0	0	0
EUR	0	283	0	6	0
AFR	0	0	14	0	0
AMR	0	0	0	27	0
SAS	0	0	0	0	2
UNK	0	0	0	0	0

(c) TCGA-OV Panel

		Inferred				
	pop	EAS	EUR	AFR	AMR	SAS
Cancer-free WES	EAS	10	0	0	0	0
	EUR	0	376	0	2	0
	AFR	0	0	28	1	0
	AMR	0	4	0	13	0
	SAS	0	0	0	0	7
	UNK	0	2	0	0	0

(d) BeatAML Panel

	Inferred				
pop	EAS	EUR	AFR	AMR	SAS
EAS	11	0	0	0	0
EUR	0	286	0	3	0
AFR	0	0	14	0	0
AMR	0	0	0	27	0
SAS	0	0	0	0	2
UNK	0	0	0	0	0

(e) TCGA-OV RNA

		Inferred				
	pop	EAS	EUR	AFR	AMR	SAS
Cancer-free WES	EAS	4	0	0	0	0
	EUR	0	242	0	0	0
	AFR	0	0	21	0	0
	AMR	1	1	0	9	0
	SAS	0	0	0	0	4
	UNK	0	1	0	0	0

(f) BeatAML RNA

	Inferred				
pop	EAS	EUR	AFR	AMR	SAS
EAS	10	0	0	0	0
EUR	0	210	0	2	0
AFR	0	0	9	0	0
AMR	0	0	0	24	0
SAS	0	0	0	0	1
UNK	0	0	0	0	0

Discussion

With this work, we introduce a systematic approach to ancestry inference from cancer-derived molecular data.
The approach is rooted in a combination of an established, extensively used PCA-based technique of ancestry inference with a central idea of inference parameter optimization using data synthesized *in silico*.
Crucially, this combination permits a statistically rigorous assessment of inference accuracy for an individual cancer-derived molecular profile, with its unique biological (e.g. cancer type) and technical (e.g., sequencing depth and quality) properties.
Synthetic data here are used as a substitute for a real-world set of molecular profiles sharing these properties and with known ground-truth genetic ancestry. It is unrealistic to expect such a real-world set to be available in all cases.
Our tests of the resulting computational methodology on a representative subset of cancer-derived data demonstrate its accurate and robust performance.
A beta version of the R package will be available at: https://github.com/KrasnitzLab/RAIDS and we plan to submit it to Bioconductor.

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